

Female Reproductive System

Section A:

Research Article: Self-directed learning. Ideal Nutrition in pregnancy and lactation.

Use library resources or open resources (eg. Google Scholar) to find a peer reviewed article on Ideal Nutrition in pregnancy and lactation. Use the following objectives as a guide to review the selected paper. Once you can answer most of these questions, then you have completed your review and are fully prepared for discussion.

- I. For the better health of the fetus and the mother, what are the ideal nutritional recommendations that as a doctor you will give?
- II. During the ultrasound examination of a pregnant patient at 6 months, the physician notices a large translucent swelling at the lumbosacral region of the fetus.

Based on your clinical knowledge, what is the congenital anomaly present in this patient and how could this anomaly have been prevented nutritionally?

Section B: Integrated Case

A 41-year-old woman comes to the emergency department with a 1-week history of light intermittent vaginal bleeding.

She says that the bleeding has not soaked through her pad. She had her first period (menarche) at age 13 and has had irregular periods ever since with cycles that last 35 to 50 days, and occasional heavy bleeding. Her last menstrual period was 53 days ago. She is currently sexually active. She is not on any form of contraception. She has no family history of bleeding disorders, cancer, or heart disease.

Her blood pressure is 100/60 mmHg, pulse is 100/min, respirations are 20/min and temperature is 37 °C (98.6F).

Physical examination shows conjunctival pallor and breast engorgement with darkening of the nipples.

Blood is collected for analysis and ultrasound is performed.

Questions

1. Is it possible that she could be menopausal, or is she pregnant? How would you confirm if this patient is pregnant?

Ans: It is possible she could have an early menopause, but prior to ruling that out, she needs to be tested for possible pregnancy. In this case, history & clinical signs are suggestive of pregnancy. Confirmation via serum pregnancy/urine pregnancy test kits. Urine or serum beta hCG. Also, ultrasound. In this case since she is almost 8 weeks pregnant in the light of history vaginal bleed.

2. Based on her last menstrual period, which week of pregnancy is she in?

Ans: 7 weeks

3. What are the physiological and anatomical changes noticed both on maternal and fetal side at the early stages of pregnancy?

Ans: Early on in pregnancy females may experience - breast tenderness, nausea, vomiting, weight gain, there may be some skin changes like pigmentation & striae.

4. Which of the hormones are predominant in the early part of pregnancy, and what supports the corpus luteum?

Ans: The 1st hormone to appear is hCG that is secreted as early as 24 hours (after implantation) which supports the corpus luteum. The corpus luteum synthesizes progesterone and estrogen. High levels of both hormones are present, but progesterone is the more predominant hormone. Also, prolactin levels start to rise throughout pregnancy.

5. What is the function of prolactin in pregnancy?

Ans: Along with estrogen and progesterone prolactin is essential for breast development to prepare the moms breast for lactation.

6. Based on her age, she wants to undergo genetic screening and other assays to rule out any form of birth defects. What are the common techniques used for this process?

Under the current recommendations, pregnant women are offered prenatal screening during the first and second trimesters of pregnancy.

I. What are the different types of non-invasive and invasive screening tests? Mention their optimal timing.

Non-Invasive	Invasive
1. Maternal Serum Screening: 16 weeks a. First trimester screen: 11 – 14 weeks b. Second trimester screen: 14 weeks	1. Amniocentesis: 15 – 18 weeks
2. Ultrasound (fetal anomaly scan): 18 weeks c. Nuchal Translucency: 11 – 14 weeks	2. Chorionic Villus Sampling: 10 – 14 weeks

II. Distinguish between first and second trimester maternal serum screening tests. Mention the findings of maternal serum screening tests found in each of the following diseases (in addition to nuchal translucency):

(a) Trisomy 21

(b) Trisomy 18

(C) Trisomy 13

(d) Neural Tube Defects

- First trimester Tests: 11 – 14 weeks include PAPP-A (Pregnancy associated plasma protein-A) and the β -hCG (human chorionic gonadotropin)
- Second trimester tests: 14 weeks include the triple test or the quad test. The triple test looks for 3 markers which are the AFP (alpha fetoprotein), estriol (unconjugated estriol), and β -hCG. The quad test includes these three markers in addition to Inhibin A.

1. Trisomy 21:

a. First trimester serum screen:

I. PAPP-A: ↓

II. Free β -hCG: ↑

b. Second trimester serum screen:

- Estriol: ↓

- AFP: ↓

- Free β -hCG: ↑

- Inhibin A: ↑

c. Nuchal Translucency: ↑

2. Trisomy 18

a. First trimester serum screen:

I. PAPP-A: ↓

II. Free β -hCG: ↓

b. Second trimester serum screen:

I. Estriol: ↓

II. AFP: ↓

III. Free β -hCG: ↓

IV. Inhibin A: N/↓

C. Nuchal Translucency: ↑

3. Trisomy 13

a. First trimester serum screen:

I. PAPP-A: ↓

II. Free β -hCG: ↑

b. Second trimester serum screen:

I. Estriol: N/↓

II. AFP: N/↓

III. Free β -hCG: N/↓

IV. Inhibin A: N

c. Nuchal Translucency: ↑

4. Neural Tube Defects:

a. Second trimester serum screen:

I. AFP: ↑

III. She is offered NIPS by next-generation sequencing of cell-free fetal DNA. Describe this test, and explain how it can identify a fetus that is at risk of having Down syndrome.

Ans: At around 7 weeks gestation, approximately 10% of the DNA found in a pregnant woman's blood is derived from the placenta of the fetus. This detectable DNA is named cell-free DNA (cfDNA). The cfDNA is used to sequence different areas of the fetus chromosome multiple times to create a statistically measurable amount of each chromosome. Surplus of any of the fetus chromosome would indicate a trisomy in the placenta.

NIPS for detecting trisomy 21 is more specific than traditional maternal serum screening (with/without nuchal translucency). It has the advantage of earlier detection of trisomy and eliminates the risk of spontaneous abortion by amniocentesis or CV sampling. Nonetheless, current guidelines recommend that a positive NIPS should be followed by CVS or amniocentesis.

IV. For a definitive karyotypic diagnosis, either Chorionic Villus Sampling (CVS) or Amniocentesis should be performed. Describe how each test is performed, the gestational timing at which it is performed, and mention one advantage and one disadvantage of CVS in comparison to Amniocentesis.

Ans: Amniocentesis involves aspiration of 10 – 20 ml of amniotic fluid under the guidance of ultrasound, and it is usually done at 15 – 18 weeks gestation.

CVS involves removal of fetal cells by aspiration from the inner surface of the placenta, and it is usually done at 10 – 14 weeks gestation.

The advantage of CVS is that if a genetic defect in the fetus is identified, because of this earlier diagnosis, it makes more options available to the parents. The parents will have more time to consider how to proceed with the pregnancy. In such a case, some parents may opt to perform an abortion prior to the end of the first trimester.

The CVS procedure is considered to carry an approximate risk of 1% of inducing a spontaneous abortion. The risk to the unborn child during amniocentesis is between 0.5 - 1%.

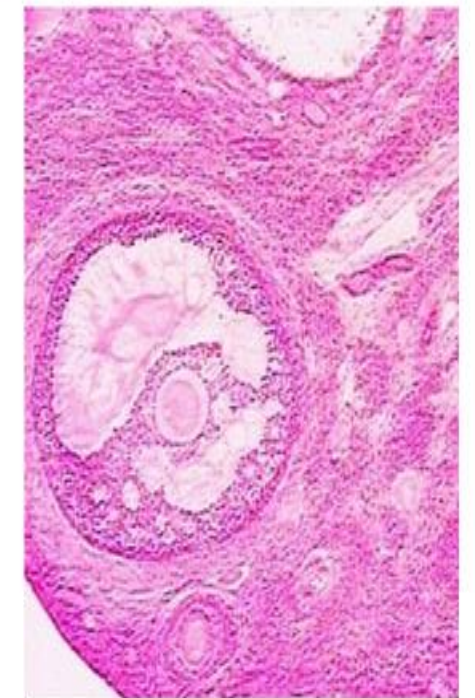
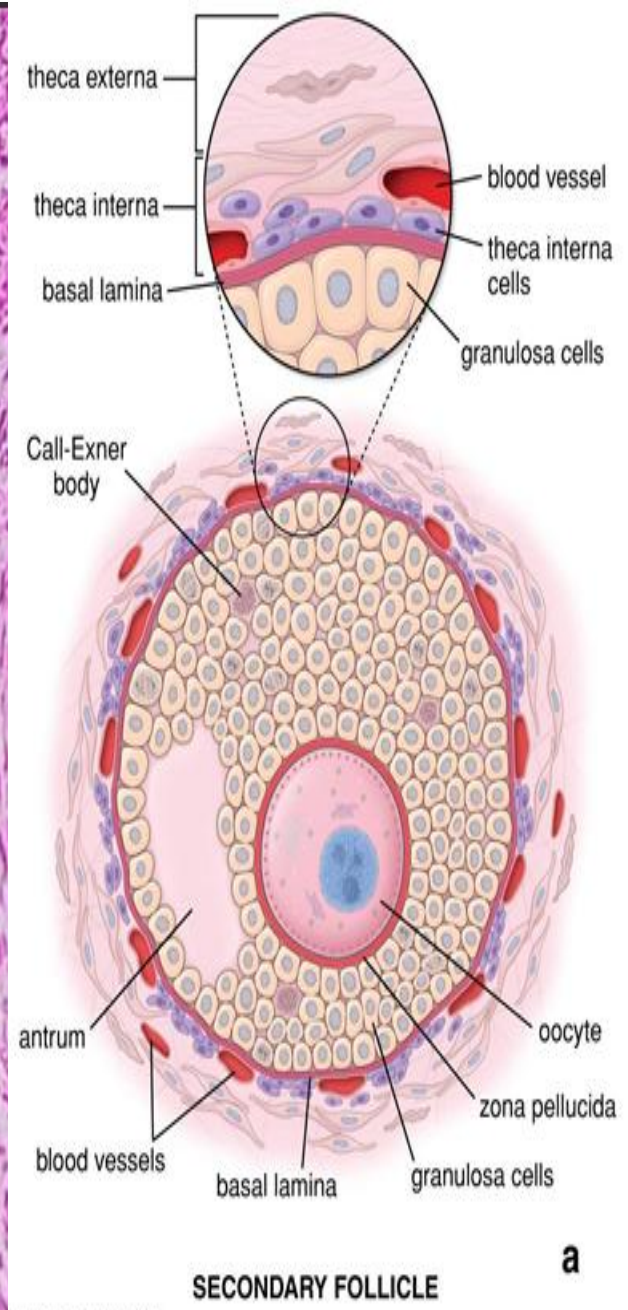
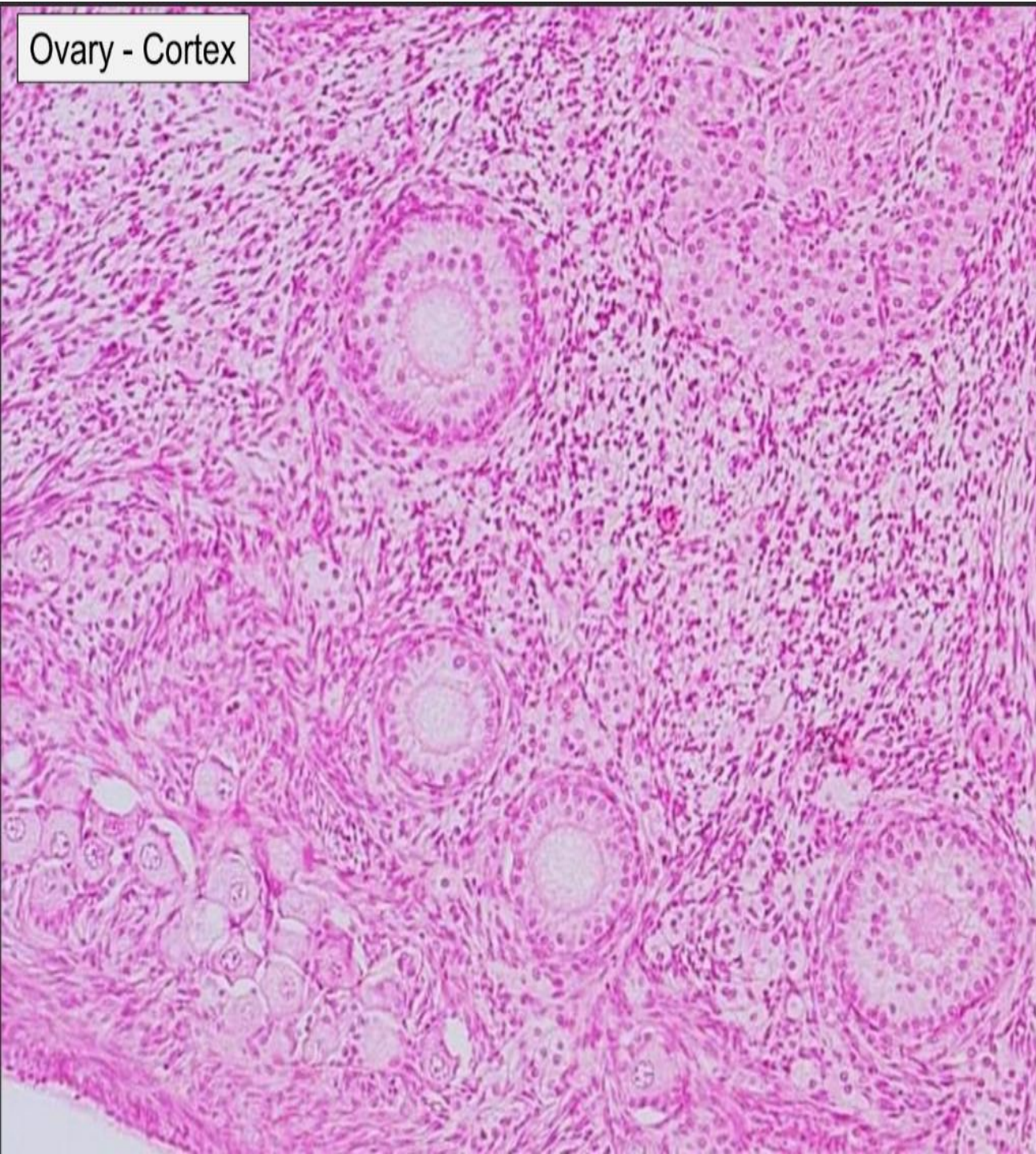
7. As we noticed the initial vaginal bleeding in this patient, which blood vessels may be responsible for this bleed?

Ans: Endometrial arteries

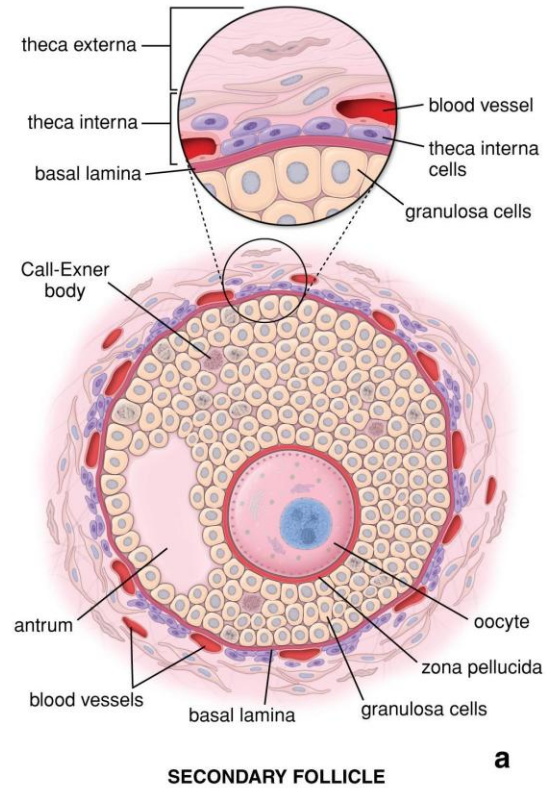
8 a. What are the changes occurring in the ovaries and uterus for the menstrual cycle to occur? Include in your discussion hormonal correlations with histological changes at the different phases (menstrual cycle, ovarian cycle and pregnancy).

Ans: Menstrual cycle, ovarian cycle and pregnancy

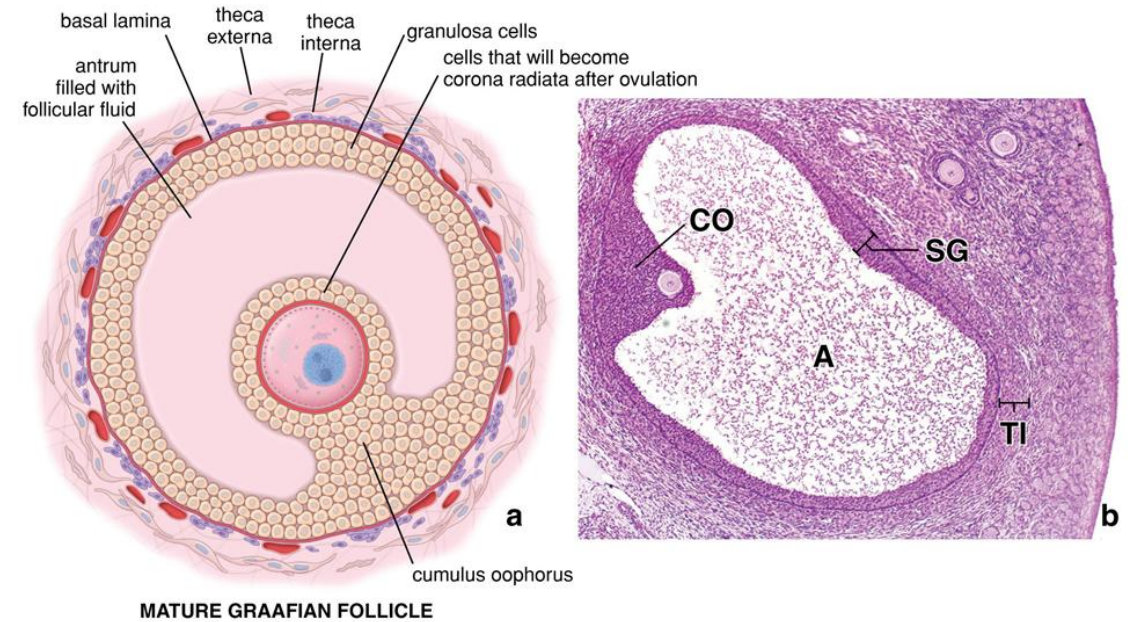
Ovary - Cortex



Ovary – Secondary Follicle

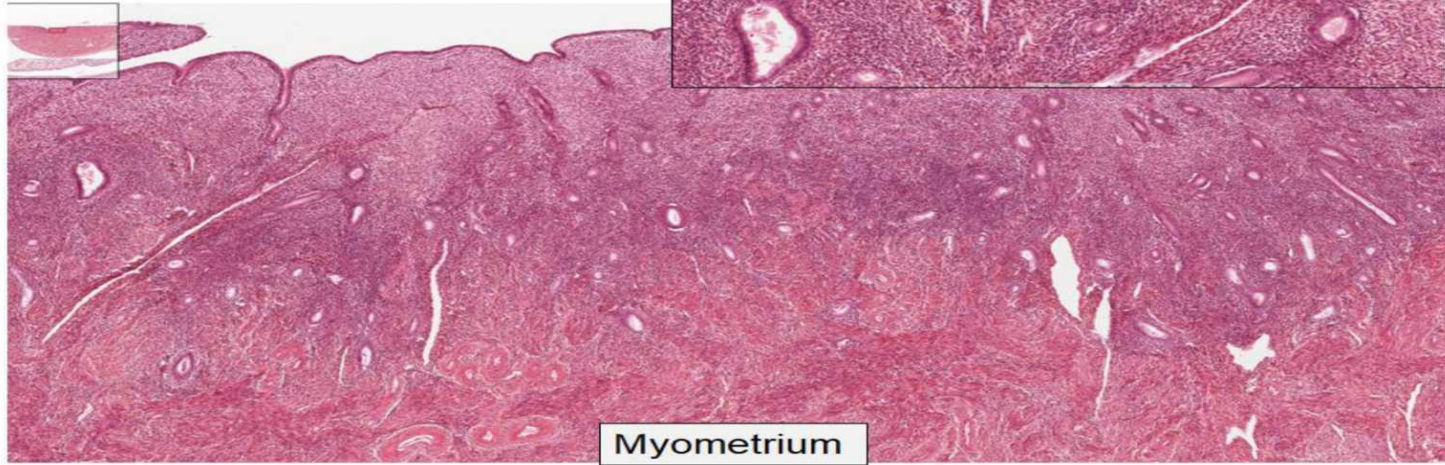


Ovary – Mature Graafian (tertiary) Follicle

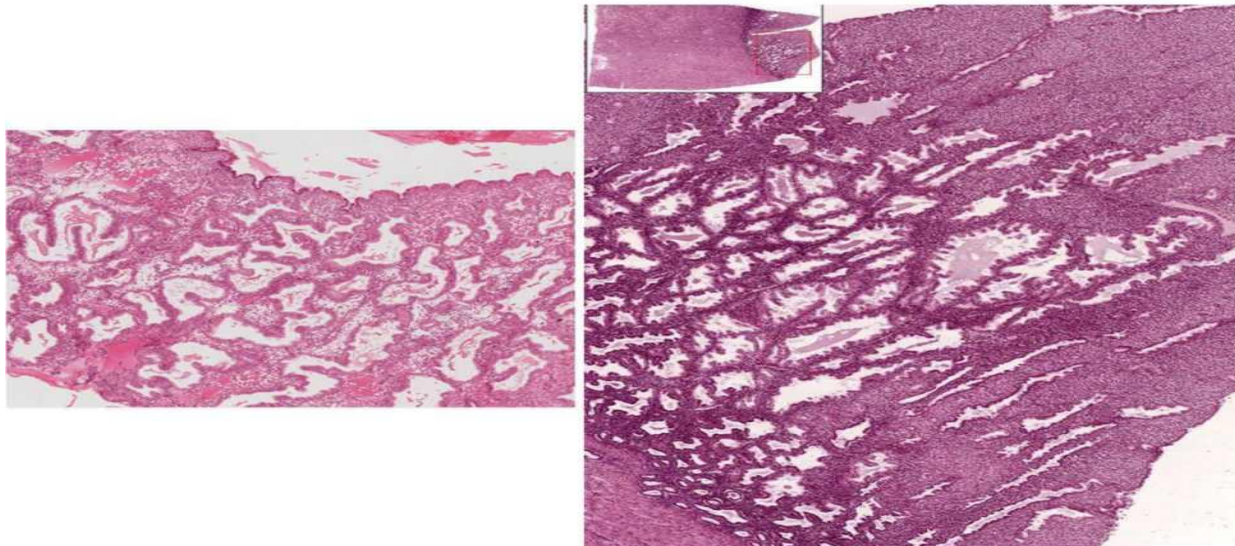




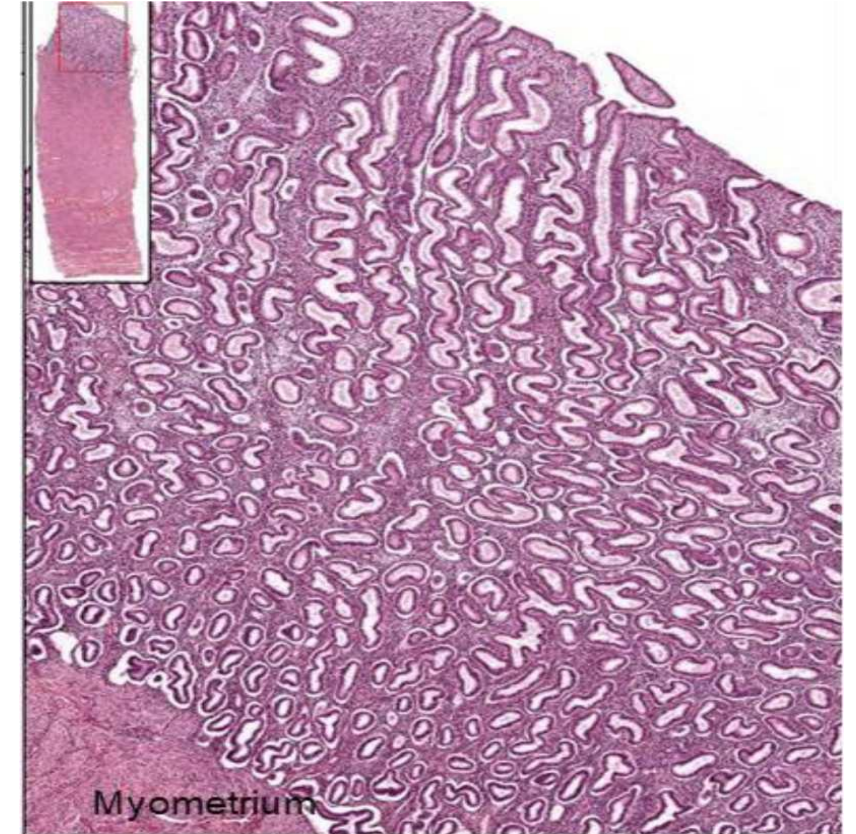
Endometrium - Proliferative Phase

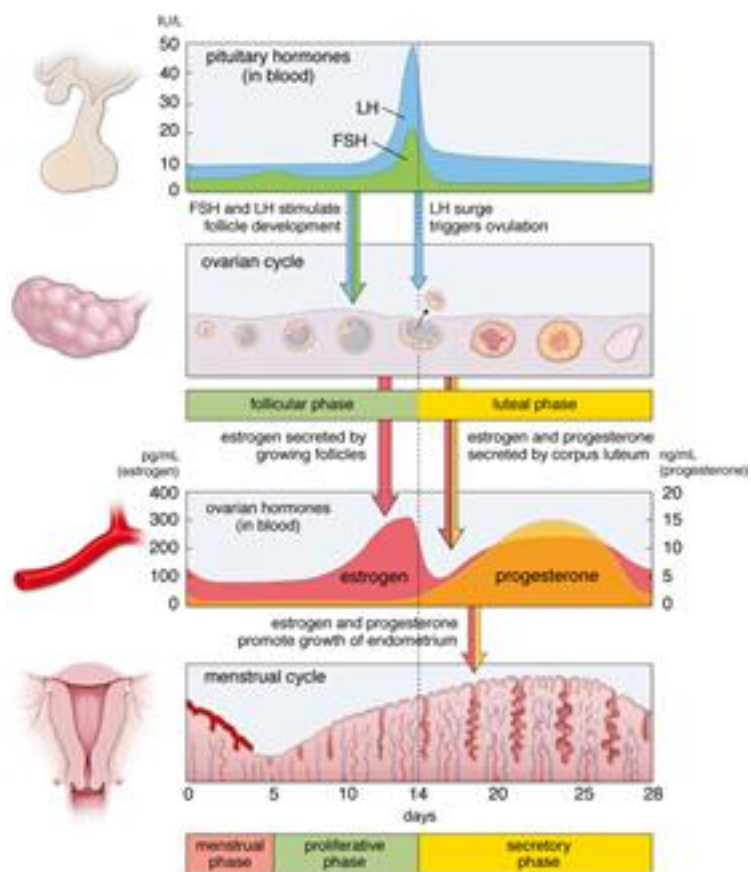


Endometrium - Late Secretory Phase



Early Secretory Phase





8 b. What changes may occur if there is pregnancy?

Ans: Endometrial changes are in preparation for implantation

Case Continued

As she attained her full-term pregnancy, she delivered at 39th week. In regard the preparing of the mother for vaginal delivery a lateral episiotomy was performed.

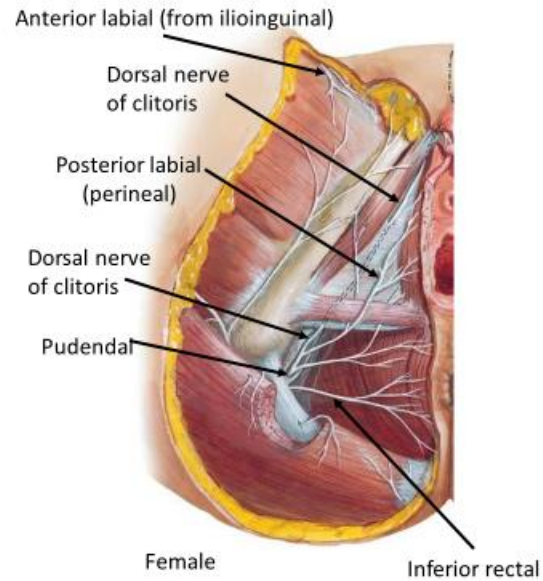
9. Discuss the innervation and lymphatic drainage of the perineum?

Ans: Lymphatic drainage:

The erectile tissues of the clitoris and the labia minora drain into the deep inguinal nodes. Lymphatics from the perineal skin drain into the superficial inguinal lymph nodes (horizontal group)

- Innervation

Nerves of the female perineum



10. What important structures are avoided by doing a lateral episiotomy?

Ans: perineal body, external anal sphincter

Case continued

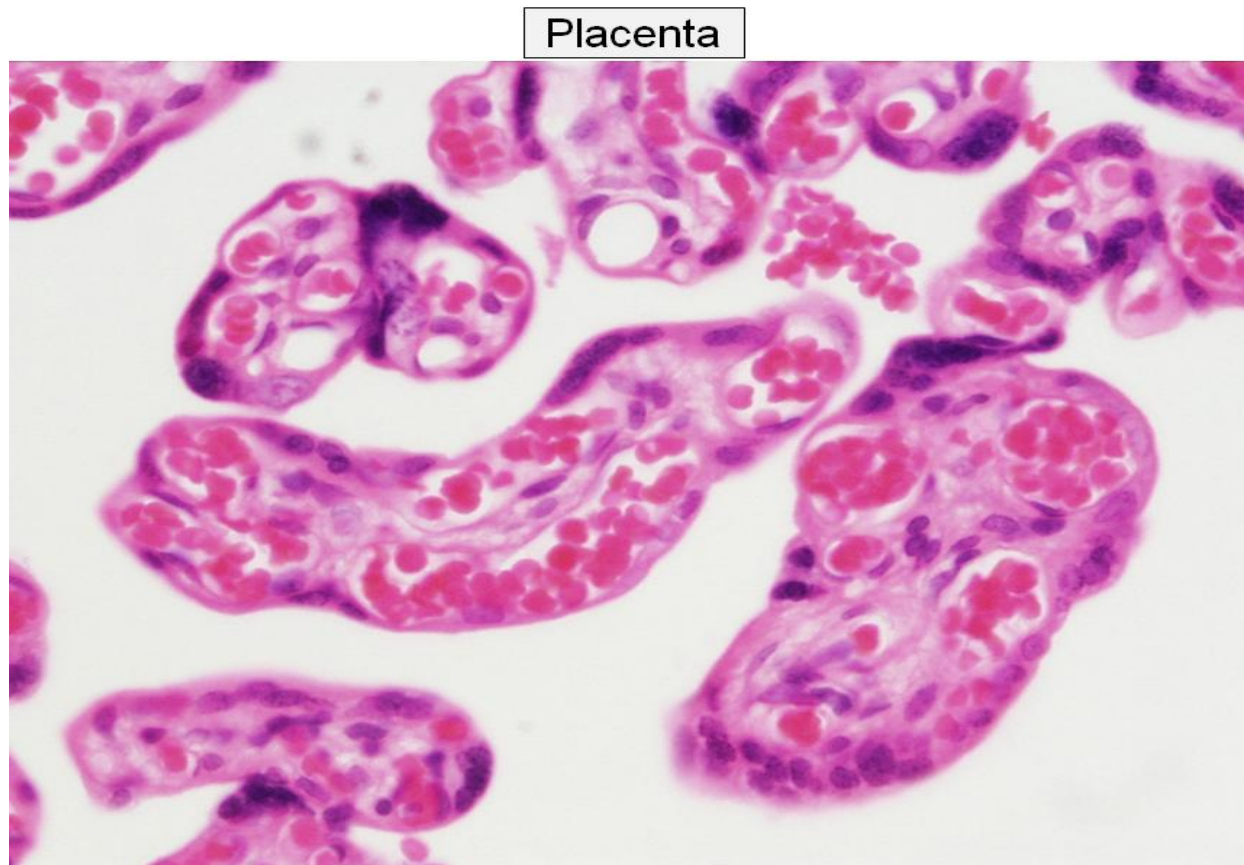
The placenta was delivered and found intact upon its examination.

11. What are the functions of placenta and discuss briefly?

Ans: Excretory, endocrine, immunologic and nutritional functions

12. Discuss briefly the different histological stages of placental development?

Ans: Primary, secondary and tertiary villus development



13. Compare the histological features of a pubertal mammary gland and a lactating mammary gland. Include hormonal influences.

